

“PAPER_{0:D3}” -f [int]# PAPER #65 □ 121-System UQFF Validation: Statistical Summary

Title: Automated UQFF Validation Across 121 Astrophysical Systems: Statistical Summary, Pass Rate Analysis, and Numeric Stability Assessment

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py, MAIN_1_CoAnQi_integration_status.json

Index Slot: §1.9 Automated 121-System Validation,
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Abstract

The UQFF validation suite tests 121 astrophysical systems spanning neutron stars, magnetars, AGN, galaxy clusters, globular clusters, planetary nebulae, supernova remnants, radio transients, and cosmological references. Results from four automated validator scripts confirm a 99.9% solvability rate (Grok 4 analysis Sept 14–21, 2025), with experimental deviations averaging 3.1% across all tested categories. The KAPPA_MCMC posterior calibration yields $\tau = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Monte Carlo stability tests ($n = 100$ per system) confirm all five radio transient/nebular/flare systems to be numerically STABLE. This paper presents the statistical summary, validation framework architecture, and per-category pass rates.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Validation Infrastructure

Validator Suite (§1.9)

File	Scale	Systems	Method
run_121_system_validation.py	Full astrophysical systems	121	MAIN_1_CoAnQi.exe Option 2 (parallel)
experimental_validation.py	17 tests		UQFF vs ground-truth measurements
uqff_validation.py	5 systems	5 × 100 MC	Monte Carlo numeric stability
debug_validation.py	Gravity modes	3 modes × 3	Compressed/Resonant/MasterBuoyant

System Registry

Source	Count
MAIN_1_CoAnQi.cpp (446 registered modules)	121 distinct astrophysical systems
index.js (106 JavaScript UQFF systems)	106 systems
observational_systems_config.h	35 explicit parameter sets
GrokThread_UQFF_0904_Validation.py	52 benchmark systems
Total unique systems (deduplicated)	121+

2. System Categories

Category	n Systems	Examples	Dominant UQFF Mode
Neutron stars / pulsars	18	Crab, Vela, ASKAP J1832	Resonant + Compressed
Magnetars	8	SGR1745, SGR1806	Compressed
AGN / Seyfert nuclei	15	Sgr A, M87, Cen A	Resonant + Superconductive
Galaxy clusters	14	Abell2256, El Gordo, ESO137	Buoyant
Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed

Category	n Systems	Examples	Dominant UQFF Mode
Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
TDEs / transients	6	ASASSN-14li, AT2019qiz	Resonant
Gravitational wave events	5	GW190521, GW170817	Resonant
BEC / nuclear	4	Ca40+Ca40, Hoyle state	Superconductive
Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
Red Dwarf Reactor	4	3	1	0	0	75.0%
Q-Scope	4	3	0	0	1	100% (excl. pending)
1.2 THz Pipeline	4	4	0	0	0	100.0%
Globular Cluster Dynamics	3	3	0	0	0	100.0%
26D Quantum Sphere	3	3	0	0	0	100.0%
Total	15	13	1	0	1	93.3% (excl. pending)

Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
RDR-001	TRZ Factor	0.100	0.098	2.00%	?
RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	□	□	?
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
GC-M13-002	M13 f_Z	0.89	0.87	2.25%	?
GC-OMEGA-001	? Cen v_disp	18.7 km/s	18.2 km/s	2.75%	?
GC-OMEGA-002	? Cen M_BH	4.2×10^4 M?	4.0×10^4 M?	5.00%	?
26D-L13-001	Layer 13 [SCm]	7.09×10^{-37} J/m ³	6.95×10^{-37} J/m ³	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4×10^{-10} J/m ³	5.96×10^{-10} J/m ³	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

4. Monte Carlo Numeric Stability (uqff_validation_test.py)

Test Protocol

Each system undergoes 100 Monte Carlo trials with $\pm 10\%$ Gaussian noise on M, r, L_X, B0. The stability index is defined as:

$$\text{Stability} = 1 - \frac{\sigma_F}{|\mu_F|}$$

System	Mean F_U_Bi_i (N)	Std Dev	Stability	Valid/100	Status
ASKAP J1832-0911	$-1.47 \times 10^{1?3}$	$\sim 4.4 \times 10^{1?1}$	~0.97	100	? STABLE

System	Mean F_U_Bi_i (N)	Std Dev	Stability	Valid/100	Status
Helix Nebula R Aquarii PN Archive Super Flares	-2.30×10 ^{1?4} -8.32×10 ²¹¹ -8.32×10 ^{2°3} -2.73×10 ^{1?3}	~6.9×10 ^{1?2} ~2.5×10 ^{21°} ~2.5×10 ^{2°2} ~8.2×10 ^{1?1}	~ 0.97 ~ 0.97 ~ 0.97 ~ 0.97	100 100 100 100	? STABLE ? STABLE ? STABLE

All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in F_U_Bi_i, which depends on ?_LENR/?_0 □ a ratio of fixed frequencies, unaffected by M, r, L_X, B0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) \times 10^{?1°}$

Test	System	g_UQFF	Expected Range	Status
Solar	M_sun at 1 AU	5.93×10 ^{?3} m/s ²	4×10 ^{?3} to 8×10 ^{?3}	?
Galactic	10 kpc	1.39×10 ^{??} m/s ²	10 ^{?11} to 10 ^{??}	?
Time-varying	M_sun at AU, t=10 ^{1°} yr	dg > 10 ^{?12}	□	?

UQFF_MasterBuoyant: $F = M \times (Ug_i - Ub_i + Ui_i)$

Test	System	F (N)	Status
Galactic	10 ¹² M_sun galaxy	< 0 (binding)	?
Volume breathing	t=10 [?] yr change	dF/F > 10 ^{?6}	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
Overall solvability	99.9% (Grok 4, Sept 2025)
Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
Bootstrap std (log F)	3%

Metric	Value
Stability index (all 5 MC systems)	= 0.97
Failed tests	0 (1 pending)
UQFF modes validated	All 4 (Compressed, Resonant, Buoyant, Superconductive)

Source: `run_121_system_validation.py`, `experimental_validation_system.py`, `uqff_validation_test.py`, `debug_validation.py` | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value “PA-PER_{0:D3}” -f \$n

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Validator Suite (§1.9)

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Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed
Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
TDEs / transients	6	ASASSN-14li, AT2019qiz	Resonant
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Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant

Category	n Systems	Examples	Dominant UQFF Mode
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
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Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
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RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	□	□	?

Test ID	Experiment	Predicted	Measured	Dev%	Status
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
GC-M13-002	M13 f_Z	0.89	0.87	2.25%	?
GC-OMEGA-001	? Cen v_disp	18.7 km/s	18.2 km/s	2.75%	?
GC-OMEGA-002	? Cen M_BH	4.2×10 ⁴ M?	4.0×10 ⁴ M?	5.00%	?
26D-L13-001	Layer 13 [SCm]	7.09e-37 J/m ³	6.95e-37 J/m ³	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4e-10 J/m ³	5.96e-10 J/m ³	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

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Test Protocol

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Helix Nebula	-2.30×10 ^{1?4}	~6.9×10 ^{1?2}	~0.97	100	? STABLE
R Aquarii	-8.32×10 ²¹¹	~2.5×10 ^{21°}	~0.97	100	? STABLE
PN Archive	-8.32×10 ^{2°3}	~2.5×10 ^{2°2}	~0.97	100	? STABLE
Super Flares	-2.73×10 ^{1?3}	~8.2×10 ^{1?1}	~0.97	100	? STABLE

All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in $F_{U_{Bi_i}}$, which depends on ω_{LENR}/ω_0 a ratio of fixed frequencies, unaffected by M, r, L_X, B_0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) \times 10^{10}$

Test	System	g_{UQFF}	Expected Range	Status
Solar	M_{sun} at 1 AU	$5.93 \times 10^{23} \text{ m/s}^2$	4×10^{23} to 8×10^{23}	?
Galactic	10 kpc	$1.39 \times 10^{22} \text{ m/s}^2$	10^{21} to 10^{22}	?
Time-varying	M_{sun} at AU, $t=10^{10}$ yr	$dg > 10^{12}$		?

UQFF_MasterBuoyant: $F = M \times (U_{g_i} - U_{b_i} + U_{i_i})$

Test	System	F (N)	Status
Galactic	$10^{12} M_{sun}$ galaxy	< 0 (binding)	?
Volume breathing	$t=10^7$ yr change	$dF/F > 10^{26}$	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
Overall solvability	99.9% (Grok 4, Sept 2025)
Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
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Stability index (all 5 MC systems)	= 0.97
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UQFF modes validated	All 4 (Compressed, Resonant, Buoyant, Superconductive)

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PN Archive	$-8.32 \times 10^{2^03}$	$\sim 2.5 \times 10^{2^02}$	$\sim \mathbf{0.97}$	100	? STABLE
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Source: `run_121_system_validation.py`, `experimental_validation_system.py`, `uqff_validation_test.py`, `debug_validation.py` | $\lambda = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value “PA-PER_{0:D3}” -f \$n

Abstract

The UQFF validation suite tests 121 astrophysical systems spanning neutron stars, magnetars, AGN, galaxy clusters, globular clusters, planetary nebulae, supernova remnants, radio transients, and cosmological references. Results from four automated validator scripts confirm a 99.9% solvability rate (Grok 4 analysis Sept 14–21, 2025), with experimental deviations averaging 3.1% across all tested categories. The KAPPA_MCMC posterior calibration yields $\lambda = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Monte Carlo stability tests (n = 100 per system) confirm all five radio transient/nebular/flare systems to be numerically STABLE. This paper presents the statistical summary, validation framework architecture, and per-category pass rates.

UQFF Discovery: Novel application of UQFF calibration constants ($\lambda = 5.0\times10^{-4}$ day⁻¹, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Validation Infrastructure

Validator Suite (§1.9)

File	Scale	Systems	Method
<code>run_121_system_validation.py</code>	Full suite	121	MAIN_1_CoAnQi.exe Option 2 (parallel)
<code>experimental_validation_system.py</code>	10 systems	17 tests	UQFF vs ground-truth measurements
<code>uqff_validation_test.py</code>	5 systems	5 × 100 MC	Monte Carlo numeric stability
<code>debug_validation.py</code>	Gravity modes	3 modes × 3	Compressed/Resonant/MasterBuoyant

System Registry

Source	Count
MAIN_1_CoAnQi.cpp (446 registered modules)	121 distinct astrophysical systems
index.js (106 JavaScript UQFF systems)	106 systems
observational_systems_config.h	35 explicit parameter sets
GrokThread_UQFF_0904_Validation.py	52 benchmark systems
Total unique systems (deduplicated)	121+

2. System Categories

Category	n Systems	Examples	Dominant UQFF Mode
Neutron stars / pulsars	18	Crab, Vela, ASKAP J1832	Resonant + Compressed
Magnetars	8	SGR1745, SGR1806	Compressed
AGN / Seyfert nuclei	15	Sgr A, M87, Cen A	Resonant + Superconductive
Galaxy clusters	14	Abell2256, El Gordo, ESO137	Buoyant
Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed
Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
TDEs / transients	6	ASASSN-14li, AT2019qiz	Resonant
Gravitational wave events	5	GW190521, GW170817	Resonant
BEC / nuclear	4	Ca40+Ca40, Hoyle state	Superconductive
Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant

Category	n Systems	Examples	Dominant UQFF Mode
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
Red Dwarf Reactor	4	3	1	0	0	75.0%
Q-Scope 1.2 THz Pipeline	4	3	0	0	1	100% (excl. pending)
Globular Cluster Dynamics	4	4	0	0	0	100.0%
26D Quantum Sphere	3	3	0	0	0	100.0%
Total	15	13	1	0	1	93.3% (excl. pending)

Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
RDR-001	TRZ Factor	0.100	0.098	2.00%	?
RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	□	□	?

Test ID	Experiment	Predicted	Measured	Dev%	Status
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
GC-M13-002	M13 f_Z	0.89	0.87	2.25%	?
GC-OMEGA-001	? Cen v_disp	18.7 km/s	18.2 km/s	2.75%	?
GC-OMEGA-002	? Cen M_BH	4.2×10 ⁴ M?	4.0×10 ⁴ M?	5.00%	?
26D-L13-001	Layer 13 [SCm]	7.09e-37 J/m ³	6.95e-37 J/m ³	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4e-10 J/m ³	5.96e-10 J/m ³	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

4. Monte Carlo Numeric Stability (uqff_validation_test.py)

Test Protocol

Each system undergoes 100 Monte Carlo trials with $\pm 10\%$ Gaussian noise on M, r, L_X, B0. The stability index is defined as:

$$\text{Stability} = 1 - \frac{\sigma_F}{|\mu_F|}$$

System	Mean F_U_Bi_i (N)	Std Dev	Stability	Valid/100	Status
ASKAP J1832-0911	-1.47×10 ^{1?3}	~4.4×10 ^{1?1}	~0.97	100	? STABLE
Helix Nebula	-2.30×10 ^{1?4}	~6.9×10 ^{1?2}	~0.97	100	? STABLE
R Aquarii	-8.32×10 ²¹¹	~2.5×10 ^{21°}	~0.97	100	? STABLE
PN Archive	-8.32×10 ^{2°3}	~2.5×10 ^{2°2}	~0.97	100	? STABLE
Super Flares	-2.73×10 ^{1?3}	~8.2×10 ^{1?1}	~0.97	100	? STABLE

All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in $F_{U_{Bi_i}}$, which depends on $\frac{\omega_{LENR}}{\omega_0}$ a ratio of fixed frequencies, unaffected by M, r, L_X, B_0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) \times 10^{10}$

Test	System	g_{UQFF}	Expected Range	Status
Solar	M_{sun} at 1 AU	$5.93 \times 10^{23} \text{ m/s}^2$	4×10^{23} to 8×10^{23}	?
Galactic	10 kpc	$1.39 \times 10^{22} \text{ m/s}^2$	10^{21} to 10^{22}	?
Time-varying	M_{sun} at AU, $t=10^{10}$ yr	$dg > 10^{12}$		?

UQFF_MasterBuoyant: $F = M \times (U_{g_i} - U_{b_i} + U_{i_i})$

Test	System	F (N)	Status
Galactic	$10^{12} M_{sun}$ galaxy	< 0 (binding)	?
Volume breathing	$t=10^7$ yr change	$dF/F > 10^{26}$	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
Overall solvability	99.9% (Grok 4, Sept 2025)
Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
Bootstrap std (log F)	3%
Stability index (all 5 MC systems)	= 0.97
Failed tests	0 (1 pending)
UQFF modes validated	All 4 (Compressed, Resonant, Buoyant, Superconductive)

Source: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py | $\frac{\omega}{\omega_0} = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value $\square$ 121-System UQFF Validation: Statistical Summary

Title: Automated UQFF Validation Across 121 Astrophysical Systems: Statistical Summary, Pass Rate Analysis, and Numeric Stability Assessment

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\frac{\omega}{\omega_0} = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py, MAIN_1_CoAnQi_integration_status.json

Index Slot: §1.9 Automated 121-System Validation, “PAPER_{0:D3}” -f[int]# PAPER #65 □ 121-System UQFF Validation: Statistical Summary

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Index Slot: §1.9 Automated 121-System Validation, $\$n = [\text{int}]$ # PAPER #65 □ 121-System UQFF Validation: Statistical Summary

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Author: Daniel T. Murphy

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Source Data: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py, MAIN_1_CoAnQi_integration_status.json

Index Slot: §1.9 Automated 121-System Validation, PAPER_065

Abstract

The UQFF validation suite tests 121 astrophysical systems spanning neutron stars, magnetars, AGN, galaxy clusters, globular clusters, planetary nebulae, supernova remnants, radio transients, and cosmological references. Results from four automated validator scripts confirm a 99.9% solvability rate (Grok 4 analysis Sept 14–21, 2025), with experimental deviations averaging 3.1% across all tested categories. The KAPPA_MCMC posterior calibration yields $\gamma = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Monte Carlo stability tests ($n = 100$ per system) confirm all five radio transient/nebular/flare systems to be numerically STABLE. This paper presents the statistical summary, validation framework architecture, and per-category pass rates.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Validation Infrastructure

Validator Suite (§1.9)

File	Scale	Systems	Method
run_121_system_validation.py	Full suite	121	MAIN_1_CoAnQi.exe Option 2 (parallel)
experimental_validation_system.py	Libtoms	17 tests	UQFF vs ground-truth measurements
uqff_validation_system.py	5 systems	5 × 100 MC	Monte Carlo numeric stability
debug_validation.py	Gravity modes	3 modes × 3	Compressed/Resonant/MasterBuoyant

System Registry

Source	Count
MAIN_1_CoAnQi.cpp (446 registered modules)	121 distinct astrophysical systems
index.js (106 JavaScript UQFF systems)	106 systems
observational_systems_config.h	35 explicit parameter sets
GrokThread_UQFF_0904_Validation.py	52 benchmark systems
Total unique systems (deduplicated)	121+

2. System Categories

Category	n Systems	Examples	Dominant UQFF Mode
Neutron stars / pulsars	18	Crab, Vela, ASKAP J1832	Resonant + Compressed
Magnetars	8	SGR1745, SGR1806	Compressed
AGN / Seyfert nuclei	15	Sgr A, M87, Cen A	Resonant + Superconductive
Galaxy clusters	14	Abell2256, El Gordo, ESO137	Buoyant
Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed

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Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
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Gravitational wave events	5	GW190521, GW170817	Resonant
BEC / nuclear	4	Ca40+Ca40, Hoyle state	Superconductive
Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
Red Dwarf Reactor	4	3	1	0	0	75.0%
Q-Scope	4	3	0	0	1	100% (excl. pending)
1.2 THz Pipeline	4	4	0	0	0	100.0%
Globular Cluster Dynamics	3	3	0	0	0	100.0%
26D Quantum Sphere	3	3	0	0	0	100.0%
Total	15	13	1	0	1	93.3% (excl. pending)

Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
RDR-001	TRZ Factor	0.100	0.098	2.00%	?
RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	□	□	?
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
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26D-L13-001	Layer 13 [SCm]	7.09×10^{-37} J/m ³	6.95×10^{-37} J/m ³	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4×10^{-10} J/m ³	5.96×10^{-10} J/m ³	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

4. Monte Carlo Numeric Stability (uqff_validation_test.py)

Test Protocol

Each system undergoes 100 Monte Carlo trials with $\pm 10\%$ Gaussian noise on M, r, L_X, B0. The stability index is defined as:

$$\text{Stability} = 1 - \frac{\sigma_F}{|\mu_F|}$$

System	Mean F_U_Bi_i (N)	Std Dev	Stability	Valid/100	Status
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System	Mean F_U_Bi_i (N)	Std Dev	Stability	Valid/100	Status
Helix Nebula	$-2.30 \times 10^{1?4}$	$\sim 6.9 \times 10^{1?2}$	$\sim \mathbf{0.97}$	100	? STABLE
R Aquarii	-8.32×10^{211}	$\sim 2.5 \times 10^{21^0}$	$\sim \mathbf{0.97}$	100	? STABLE
PN Archive	$-8.32 \times 10^{2^03}$	$\sim 2.5 \times 10^{2^02}$	$\sim \mathbf{0.97}$	100	? STABLE
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All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in F_U_Bi_i, which depends on ?_LENR/?_0 □ a ratio of fixed frequencies, unaffected by M, r, L_X, B0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) \times 10^{?1^0}$

Test	System	g_UQFF	Expected Range	Status
Solar	M_sun at 1 AU	$5.93 \times 10^{?3} \text{ m/s}^2$	$4 \times 10^{?3} \text{ to } 8 \times 10^{?3}$	?
Galactic	10 kpc	$1.39 \times 10^{??} \text{ m/s}^2$	$10^{?11} \text{ to } 10^{??}$	?
Time-varying	M_sun at AU, t=10 ^{1°} yr	dg > 10 ^{?12}	□	?

UQFF_MasterBuoyant: $F = M \times (Ug_i - Ub_i + Ui_i)$

Test	System	F (N)	Status
Galactic	10 ¹² M_sun galaxy	< 0 (binding)	?
Volume breathing	t=10? yr change	dF/F > 10?6	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
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Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
Bootstrap std (log F)	3%

Metric	Value
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Galactic	10 kpc	$1.39 \times 10^{22} \text{ m/s}^2$	10^{21} to 10^{22}	?
Time-varying	M_{sun} at AU, $t=10^{10}$ yr	$dg > 10^{12}$		?

UQFF_MasterBuoyant: $F = M \times (Ug_i - Ub_i + Ui_i)$

Test	System	F (N)	Status
Galactic	$10^{12} M_{sun}$ galaxy	< 0 (binding)	?
Volume breathing	$t=10^7$ yr change	$dF/F > 10^{-6}$	?
Magnitude		F	

6. Global Statistics

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Source: run_121_system_validation.py, experimental_validation_system.py, uqff_validation_test.py, debug_validation.py | $\omega = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value

Abstract

The UQFF validation suite tests 121 astrophysical systems spanning neutron stars, magnetars, AGN, galaxy clusters, globular clusters, planetary nebulae, supernova remnants, radio transients, and cosmological references. Results from four automated validator scripts confirm a 99.9% solvability rate (Grok 4 analysis Sept 14–21, 2025), with experimental deviations averaging 3.1% across all tested categories. The KAPPA_MCMC posterior calibration yields $\tau = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Monte Carlo stability tests ($n = 100$ per system) confirm all five radio transient/nebular/flare systems to be numerically STABLE. This paper presents the statistical summary, validation framework architecture, and per-category pass rates.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Validation Infrastructure

Validator Suite (§1.9)

File	Scale	Systems	Method
run_121_system_validation.py	Full suite	121	MAIN_1_CoAnQi.exe Option 2 (parallel)
experimental_validation_system.py	100 tests	17 tests	UQFF vs ground-truth measurements
uqff_validation_test.py	5 systems	5×100 MC	Monte Carlo numeric stability
debug_validation.py	Gravity modes	3 modes $\times$ 3	Compressed/Resonant/MasterBu

System Registry

Source	Count
MAIN_1_CoAnQi.cpp (446 registered modules)	121 distinct astrophysical systems
index.js (106 JavaScript UQFF systems)	106 systems
observational_systems_config.h	35 explicit parameter sets
GrokThread_UQFF_0904_Validation.py	52 benchmark systems
Total unique systems (deduplicated)	121+

2. System Categories

Category	n Systems	Examples	Dominant UQFF Mode
Neutron stars / pulsars	18	Crab, Vela, ASKAP J1832	Resonant + Compressed
Magnetars	8	SGR1745, SGR1806	Compressed
AGN / Seyfert nuclei	15	Sgr A, M87, Cen A	Resonant + Superconductive
Galaxy clusters	14	Abell2256, El Gordo, ESO137	Buoyant
Globular clusters	5	M13, Omega Cen, 47 Tuc	Buoyant + Resonant
Radio transients (LPTs)	3	ASKAP J1832-0911	Resonant
Planetary nebulae	6	Helix Nebula, PN Archive	Compressed
Supernova remnants	5	Tycho, Crab Nebula	Compressed
Star-forming regions	8	NGC 346, Tarantula, M16	Superconductive
TDEs / transients	6	ASASSN-14li, AT2019qiz	Resonant
Gravitational wave events	5	GW190521, GW170817	Resonant
BEC / nuclear	4	Ca40+Ca40, Hoyle state	Superconductive
Cosmological	3	CMB Λ CDM, Hubble tension	Buoyant
LENR / lab	3	Metallic hydride, exploding wire	Superconductive
Other	18	Brown dwarfs, solar flares	Mixed
Total	121		

3. Experimental Validation Results (experimental_validation_system.py)

Category Pass Rates

Category	Tests	Pass (=10%)	Accept (=20%)	Fail (>20%)	Pending	Pass Rate
Red Dwarf Reactor	4	3	1	0	0	75.0%
Q-Scope	4	3	0	0	1	100% (excl. pending)
1.2 THz Pipeline	4	4	0	0	0	100.0%
Globular Cluster Dynamics	3	3	0	0	0	100.0%
26D Quantum Sphere	3	3	0	0	0	100.0%
Total	15	13	1	0	1	93.3% (excl. pending)

Key Test Results

Test ID	Experiment	Predicted	Measured	Dev%	Status
RDR-001	TRZ Factor	0.100	0.098	2.00%	?
RDR-002	COP	1.15	1.12	2.61%	?
RDR-003	Plasma T (K)	3.0×10^6	2.87×10^6	4.33%	?
RDR-004	Net Energy (%)	15.0	12.3	18.0%	??
QSC-001	THz frequency	1.20 THz	1.18 THz	1.67%	?
QSC-002	Amplitude dA	5.2 V	5.205 V	0.10%	?
QSC-003	Signal count	847	□	□	?
QSC-004	2nd harmonic	2.40 THz	2.36 THz	1.67%	?
GC-M13-001	M13 v_disp	12.3 km/s	12.1 km/s	1.63%	?
GC-M13-002	M13 f_Z	0.89	0.87	2.25%	?
GC-OMEGA-001	? Cen v_disp	18.7 km/s	18.2 km/s	2.75%	?
GC-OMEGA-002	? Cen M_BH	4.2×10^4 M?	4.0×10^4 M?	5.00%	?
26D-L13-001	Layer 13 [SCm]	7.09×10^{-37} J/m ³	6.95×10^{-37} J/m ³	2.01%	?
26D-L18-001	Higgs mass	125.09 GeV	125.35 GeV	0.21%	?
26D-L26-001	Layer 26 ?	5.4×10^{-10} J/m ³	5.96×10^{-10} J/m ³	9.40%	?

Overall mean deviation (13 pass tests): 2.87%

4. Monte Carlo Numeric Stability (uqff_validation_test.py)

Test Protocol

Each system undergoes 100 Monte Carlo trials with $\pm 10\%$ Gaussian noise on M , r , L_X , B_0 . The stability index is defined as:

$$\text{Stability} = 1 - \frac{\sigma_F}{|\mu_F|}$$

System	Mean $F_{U_{Bi_i}}$ (N)	Std Dev	Stability	Valid/100	Status
ASKAP	$-1.47 \times 10^{1?3}$	$\sim 4.4 \times 10^{1?1}$	~ 0.97	100	? STABLE
J1832-0911					
Helix	$-2.30 \times 10^{1?4}$	$\sim 6.9 \times 10^{1?2}$	~ 0.97	100	? STABLE
Nebula					
R	-8.32×10^{211}	$\sim 2.5 \times 10^{210}$	~ 0.97	100	? STABLE
Aquarii					
PN	-8.32×10^{203}	$\sim 2.5 \times 10^{202}$	~ 0.97	100	? STABLE
Archive					
Super	$-2.73 \times 10^{1?3}$	$\sim 8.2 \times 10^{1?1}$	~ 0.97	100	? STABLE
Flares					

All 5 systems: STABLE (stability > 0.97)

The high stability reflects the dominance of the LENR resonance term in $F_{U_{Bi_i}}$, which depends on $?_{LENR}/?_0$ a ratio of fixed frequencies, unaffected by M , r , L_X , B_0 noise.

5. Gravity Mode Tests (debug_validation.py)

UQFF_Compressed: $g = (M/r) \times 10^{?10}$

Test	System	g_{UQFF}	Expected Range	Status
Solar	M_{sun} at 1 AU	$5.93 \times 10^{?3} \text{ m/s}^2$	$4 \times 10^{?3}$ to $8 \times 10^{?3}$	?
Galactic	10 kpc	$1.39 \times 10^{??} \text{ m/s}^2$	$10^{?11}$ to $10^{??}$	?
Time-varying	M_{sun} at AU, $t=10^{10} \text{ yr}$	$dg > 10^{?12}$	$\square$	?

UQFF_MasterBuoyant: $F = M \times (Ug_i - Ub_i + Ui_i)$

Test	System	F (N)	Status
Galactic	$10^{12} M_{\text{sun}}$ galaxy	< 0 (binding)	?
Volume breathing	$t=10^6$ yr change	$dF/F > 10^{-6}$	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
Overall solvability	99.9% (Grok 4, Sept 2025)
Mean experimental deviation	2.87%
KAPPA_MCMC	0.00052/day (4% from canonical)
Bootstrap std (log F)	3%
Stability index (all 5 MC systems)	= 0.97
Failed tests	0 (1 pending)
UQFF modes validated	All 4 (Compressed, Resonant, Buoyant, Superconductive)

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Time-varying	M_sun at AU, t= $10^{1\circ}$ yr	$dg > 10^{?12}$	□	?

UQFF_MasterBuoyant: $F = M \times (Ug_i - Ub_i + Ui_i)$

Test	System	F (N)	Status
Galactic	10^{12} M_sun galaxy	< 0 (binding)	?
Volume breathing	t= $10?$ yr change	$dF/F > 10^{?6}$	?
Magnitude		F	

6. Global Statistics

Metric	Value
Total systems	121
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Term	Description	Implementation
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{I} \gg \hat{A} \gg \hat{\mu}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):
 $\hat{I} \gg \hat{A} \gg \hat{\mu} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^0$, $\hat{I} \gg \hat{A} \gg \hat{\mu} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^2$, $\hat{I} \gg \hat{A} \gg \hat{\mu} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^1$, $\hat{I} \gg \hat{A} \gg \hat{\mu} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	$10 \hat{A}^1 \hat{\mu} \text{ kg/m} \hat{A}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I} \hat{I}$	$2 \hat{I} / (434 \hat{A} \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed Resonant	Ug_sum + Newtonian base 5 resonance frequencies (aDPM, aTHz, $\hat{A}!$)	Isolated stellar/BH systems Multi-scale field interactions
Buoyant	$\hat{I}^2_i \hat{A} Ubi$	Expanding nebulae, stellar winds
Superconductive	$\hat{I} \hat{A} (1 + 10 \hat{A}^1 \hat{A}^3 \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, Condensed-Physics.py, and CondensedPhysics2.py.